

General Notes:

Column visibility is largely specific to the needs of each standard

a. Maintained Illuminance Target values are consensus recommendations for minimum (Min), average (Avg), or maximum (Max) maintained illuminance levels at heights above finish floor (AFF) for application tasks/areas described in the table, for normally sighted people under 65 years of age. This table should be used in conjunction with other relevant design considerations following these governing criteria:

i. In cases where the height of the visual task may vary, the abbreviation TS (for "task surface") is used. The illuminance criteria then apply at the height of the visual task.

ii. Light loss factors such as luminaire dirt depreciation, lumen depreciation, reduced room surface and furniture reflectances, and other design criteria should be used in calculating maintained illuminances. Refer to ANSI/IES LS-6, Lighting Practice: Calculation of Light and Its Effects (see Foreword) and ANSI/IES/NALMCO RP-36, Recommended Practice for Lighting Maintenance for additional information.

iii. Illuminance levels are designated for Tasks or Areas. "T" (task) designates a visual task, and illumination level refers to that of the localized task areas, such as a desk or reading surface (horizontal illuminance), or library stacks (vertical illuminance); the illuminance recommendations apply to that task area only, through the total contribution of general and task lighting. "A" (area) designates an area or room where the lighting criterion applies to an entire area under consideration, such as a corridor floor or the field of play in a soccer field.

iv. When multiple tasks are performed within the same space, tasks should be ranked by importance, prevalence, or frequency using data that may be available from the client. From that information, the most commonly occurring task with the highest recommended illuminance should govern the illuminance recommendation for the task or area. Localized task lighting can be used for instances when additional illumination is necessary for infrequent, but more visually demanding, tasks.

b. Variances to Maintained Illuminance Target Values:

i. Health code and safety code requirements supersede these recommendations.

ii. When safety and security or human-vehicular proximity are significant concerns, recommended values are to be minimum maintained illuminances for the task area. For more guidance in applications when security is a concern, refer to IES G-1, Security Lighting for People, Property, and Critical Infrastructure.

iii. When a majority of the occupants of a space are over 65, the illuminance recommendations should be doubled. For special considerations for seniors or visually impaired people of any age, refer to ANSI/IES RP-28, Recommended Practice: Lighting and the Visual Environment for Older Adults and the Visually Impaired (see Foreword). Localized additional task lighting should be considered for occupants who may require additional lighting, before selecting higher illuminance criteria for the entire space or group.

iv. For visual tasks categorized as P through Y and under lighting conditions where the light source S/P Value is different than 1.0, variances are allowed in accordance with ANSI/IES TM-24, Technical Memorandum: An Optional Method for Adjusting the Recommended Illuminance for Visually Demanding Tasks within IES Illuminance Categories P through Y Based on Light Source Spectrum.

Acceptable tolerances for lighting calculations during the design process are within $\pm 10\%$. If a predicted value is below a recommendation by more than 10%, then a significant percentage of the users of the system may not find the visibility acceptable. If a predicted value exceeds a recommendation by more than 10%, then over-lighting and energy misuse may result.

Header	Data type	Definition/Notes
Sub Category	Hierarchy	Application Category (level 1) - Notes "INTERIOR" (dark blue fill) or "EXTERIOR" (dark green fill) - Establishes hierarchy, but never contains "data" in this row
		Application Category (level 2) - Color-coded interior (light blue fill) or exterior (dark red fill) - Establishes hierarchy, but never contains "data" in this row
App	Hierarchy	
App_s1	Hierarchy	Application Category (level 3) - Establishes hierarchy, and <i>might</i> contain "data" in this row
		Application Category (level 4) - Establishes hierarchy, and <i>might</i> contain "data" in this row - Could be unused (hierarchy does not always extend this deep)
App_s2	Hierarchy	
App_s3	Hierarchy	Application Category (level 5) - Establishes hierarchy, and <i>might</i> contain "data" in this row - Could be unused (hierarchy does not always extend this deep)

App_s4	Hierarchy	Application Category (level 6) - Establishes hierarchy, and <i>might</i> contain "data" in this row" - Could be unused (hierarchy does not always extend this deep)
App_s5	Hierarchy	Application Category (level 7) - Establishes hierarchy, and <i>might</i> contain "data" in this row" - Could be unused (hierarchy does not always extend this deep)
App_s6	Hierarchy	Application Category (level 8) - Establishes hierarchy, and <i>might</i> contain "data" in this row" - Could be unused (hierarchy does not always extend this deep)
Link Text	Hyperlinked Reference	Hyperlinked reference to another standard - Staff will add hyperlink in the PDF
T (Task)A (Area)	Application Notes: General	Informs whether the illuminance target is an "Area" (A) or "Task" (T) Informs whether there is a "High" (H), "Medium" (M) or "Low" (L) rating for veiling reflection concern.
Veilling RiskL (Low)M (Med)H (High)	Application Notes: General	Veiling reflections are luminous reflections from specular or semi-specular surfaces that physically change the contrast of the visual task, and therefore change the stimulus presented to the visual system.
Class of Play in sports is rated as category "I," "II," "III" or "IV." Consensus illuminance recommendations for Level "I" are generally the highest.		
As the skill and performance of players increase, speed also increases, which in turn calls for higher illuminance levels and/or higher contrasts between the target and the background. In general, players and spectators require a better and more sophisticated luminous environment. Some correlation exists between the size of a facility and the level of play; for example, a higher skill level attracts a greater number of spectators. In addition, a higher skill level may also have faster play, requiring increased illumination levels for the players, spectators, and broadcasting.		
(RP-6) Class of Play	Application Notes: General	
(Staff edits only) Horiz Cat	Application Notes: Horizontal Illuminance	CAT stands for “Category,” which refers to letter codes A through Y, found in ANSI/IES RP-10, Table A-2, which in turn correlate with specific illuminance targets. These categories are sometimes used as reference markers in other IES standards; however, the illuminance values within the tables themselves are the values that should be used. In some cases, the recommended maintained illuminance target values do not align with a specific category, in which case there will be blanks or dashes in the CAT column.

Consensus recommendation for illuminance level on a horizontal plane, in Lux (lx).

An approximate lux-to-footcandle conversion factor of 10:1 is used in these tables, instead of the more accurate conversion factor of 10.76:1. Acceptable tolerances for lighting calculations during the design process are within $\pm 10\%$ of the target value. If a predicted value is below a target recommendation by more than 10%, then a significant percentage of the users of the system may not find the visibility acceptable. If a predicted value exceeds a target recommendation by more than 10%, then over-lighting and energy misuse may result.

HorizLux	Application Notes: Horizontal Illuminance	
		- Numeric value indicates the height, in meteres, above a finished floor (AFF) that the horizontal illuminance measurements should be measured. - "TS" indicates that illuminance measurements along the horizontal plane should be measured at the Task Surface (TS). Task Surface means that the light level recommendation applies to whatever height the visual task is located at, rather than at any prescribed height above the floor. For instance, tables have varying heights, and the horizontal illuminance recommendation might be at TS to account for the fact that tables do not have fixed heights.
@ Meters	Application Notes: Horizontal Illuminance	Consensus recommendation for illuminance level on a horizontal plane, in footcandles.
Horiz Fc	Application Notes: Horizontal Illuminance	
		- Numeric value indicates the height, in feet, above a finished floor (AFF) that the horizontal illuminance measurements should be measured. - "TS" indicates that illuminance measurements along the horizontal plane should be measured at the Task Surface (TS). Task Surface means that the light level recommendation applies to whatever height the visual task is located at, rather than at any prescribed height above the floor. For instance, tables have varying heights, and the horizontal illuminance recommendation might be at TS to account for the fact that tables do not have fixed heights.
@ Feet	Application Notes: Horizontal Illuminance	

- "Max" = consensus recommendation for maximum maintained illuminance levels at defined height above finished floor (AFF) for the application task/area noted, for normally sighted people under 65 years of age.

- "Min" = consensus recommendation for minimum maintained illuminance levels at defined height above finished floor (AFF) for the application task/area noted, for normally sighted people under 65 years of age.

- "Avg" = consensus recommendation for average maintained illuminance levels at defined height above finished floor (AFF) for the application task/area noted, for normally sighted people under 65 years of age.

MaxAvgMin Application Notes: Horizontal Illuminance

Coefficient of Variation (CV) is a measure of the weighted average of all values and is commonly used in statistics. The variation of a set of values is calculated by the ratio of the standard deviation to the mean.

(RP-6)CV Application Notes: Horizontal Illuminance

CV is directly related to uniformity ratios. However, it does not rely solely on a single measurement point but includes every measurement point of the array. Uniformity Ratio (UR) is the most common measure of uniformity. UR is a measure for the quality of the illumination. It is defined as the ratio of the highest to the lowest calculated or measured illuminance values.

Uniformity RatioFormat 2:1 as"2 " 1" Application Notes: Horizontal Illuminance

Design criteria:

- Max:Avg:Min = ratio of the maximum value in the calculation plane to the average value in the calculation plane to the minimum value in the calculation plane
- Max:Avg = ratio of the maximum value in the calculation plane to average of values in the calculation plane
- Max:Min = ratio of the maximum value in the calculation plane to the minimum value in the calculation plane
- Avg:Min = ratio of the average values in the calculation plane to the minimum value in the calculation plane

Ratio BasisMax:Avg:MinMax:AvgMax:MinAvg:Min Application Notes: Horizontal Illuminance

CAT stands for "Category," which refers to letter codes A through Y, found in ANSI/IES RP-10, Table A-2, which in turn correlate with specific illuminance targets. These categories are sometimes used as reference markers in other IES standards; however, the illuminance values within the tables themselves are the values that should be used. In some cases, the recommended maintained illuminance target values do not align with a specific category, in which case there will be blanks or dashes in the CAT column.

(Staff edits only) Vert Cat Application Notes: Vertical Illuminance

VertLux	Application Notes: Vertical Illuminance	Consensus recommendation for illuminance level on a vertical plane, in Lux (lx). An approximate lux-to-footcandle conversion factor of 10:1 is used in these tables, instead of the more accurate conversion factor of 10.76:1. Acceptable tolerances for lighting calculations during the design process are within $\pm 10\%$ of the target value. If a predicted value is below a target recommendation by more than 10%, then a significant percentage of the users of the system may not find the visibility acceptable. If a predicted value exceeds a target recommendation by more than 10%, then over-lighting and energy misuse may result. - Numeric value indicates the height, in feet, above a finished floor (AFF) that the vertical illuminance measurements should be measured. - "TS" indicates that illuminance measurements along the vertical plane should be measured at the Task Surface (TS). Task Surface means that the light level recommendation applies to whatever height the visual task is located at, rather than at any prescribed height above the floor. For instance, tables have varying heights, and the horizontal illuminance recommendation might be at TS to account for the fact that tables do not have fixed heights.
@ Meters3 Vert Fc	Application Notes: Vertical Illuminance Application Notes: Vertical Illuminance	- Numeric value indicates the height, in feet, above a finished floor (AFF) that the vertical illuminance measurements should be measured. - "TS" indicates that illuminance measurements along the vertical plane should be measured at the Task Surface (TS). Task Surface means that the light level recommendation applies to whatever height the visual task is located at, rather than at any prescribed height above the floor. For instance, tables have varying heights, and the horizontal illuminance recommendation might be at TS to account for the fact that tables do not have fixed heights.
@ Feet4	Application Notes: Vertical Illuminance	- "Max" = consensus recommendation for maximum maintained illuminance levels at defined height above finished floor (AFF) for the application task/area noted, for normally sighted people under 65 years of age. - "Min" = consensus recommendation for minimum maintained illuminance levels at defined height above finished floor (AFF) for the application task/area noted, for normally sighted people under 65 years of age. - "Avg" = consensus recommendation for average maintained illuminance levels at defined height above finished floor (AFF) for the application task/area noted, for normally sighted people under 65 years of age.
MaxAvgMin5	Application Notes: Vertical Illuminance	

		Coefficient of Variation (CV) is a measure of the weighted average of all values and is commonly used in statistics. The variation of a set of values is calculated by the ratio of the standard deviation to the mean.
(RP-6)CV6	Application Notes: Vertical Illuminance	CV is directly related to uniformity ratios. However, it does not rely solely on a single measurement point but includes every measurement point of the array.
Uniformity Ratio	Application Notes: Vertical Illuminance	Uniformity Ratio (UR) is the most common measure of uniformity. UR is a measure for the quality of the illumination. It is defined as the ratio of the highest to the lowest calculated or measured illuminance values.
Format 2:1 as "2 " 1"7		Design criteria: - Max:Avg:Min = ratio of the maximum value in the calculation plane to the average value in the calculation plane to the minimum value in the calculation plane - Max:Avg = ratio of the maximum value in the calculation plane to average of values in the calculation plane - Max:Min = ratio of the maximum value in the calculation plane to the minimum value in the calculation plane - Avg:Min = ratio of the average values in the calculation plane to the minimum value in the calculation plane
Ratio Basis	Application Notes: Vertical Illuminance	
Max:Avg:Min	Application Notes: Environmental and Visual Considerations	
Max:Avg	- NOTE: Currently only found in RP-43-25. These columns may eventually be used by other standards, with slightly different definitions	Maximum glare rating. See application notes for definition.
Max:Min	Application Notes: Environmental and Visual Considerations	
Avg:Min	- NOTE: Currently only found in RP-43-25. These columns may eventually be used by other standards, with slightly different definitions	Maximum uplight rating. See application notes for definition.
8	Application Notes: Environmental and Visual Considerations	
Glare(max)	- NOTE: Currently only found in RP-43-25. These columns may eventually be used by other standards, with slightly different definitions	
	Application Notes: Environmental and Visual Considerations	
Uplight (max)	- NOTE: Currently only found in RP-43-25. These columns may eventually be used by other standards, with slightly different definitions	
	Application Notes: Environmental and Visual Considerations	
Controls	- NOTE: Currently only found in RP-43-25. These columns may eventually be used by other standards, with slightly different definitions	Lighting controls guidance. See application notes for definition.
	Application Notes: Environmental and Visual Considerations	
Spectrum	- NOTE: Currently only found in RP-43-25. These columns may eventually be used by other standards, with slightly different definitions	Lighting spectrum guidance. See application notes for definition.
Indoor/Outdoor?	(Denoted by Sub Category text, plus Sub Category and App fill colors)	See notes for "Sub Category" and "App"